YEAR 9 MATHS · CHAPTER 1 · MIND MAP

Number and financial maths

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1 Integer operations

Multiply or divide: same signs give +, different signs give −.

$4 + (-9) = -5 \quad -6 - (-2) = -4$

$-7 \times (-3) = 21 \quad -24 \div 6 = -4$

Order of operations: $-2 + 3 \times (-4) = -14$

Watch out: $(-3)^2 = 9$ but $-3^2 = -9$

2 Decimal places and significant figures

Look at the next digit. 5 or more rounds up; 4 or less stays.

3.476 to 1 d.p. is **3.5**, to 2 d.p. is **3.48**

0.04082 to 2 s.f. is **0.041** (start at the first non-zero digit)

5683 to 1 s.f. is **6000**

3 Rational numbers

Rational means it can be written as a fraction $\frac{a}{b}$.

$2\frac{2}{5} = \frac{12}{5} = 2.4$ (terminating)

$\frac{2}{3} = 0.666\ldots = 0.\dot{6}$ (recurring)

$0.35 = \frac{35}{100} = \frac{7}{20}$

$\sqrt{2}$ and π are irrational: their decimals never end or repeat.

4 Operations with fractions

Add or subtract: common denominator first. Divide: flip the second fraction and multiply.

$2\frac{1}{3} + 1\frac{3}{4} = \frac{28}{12} + \frac{21}{12} = \frac{49}{12} = 4\frac{1}{12}$

$\frac{3}{5} \div \frac{9}{10} = \frac{3}{5} \times \frac{10}{9} = \frac{2}{3}$

Watch out: $\frac{1}{2} + \frac{1}{3} \neq \frac{2}{5}$. It's $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

5 Ratios, rates and best buys

Simplify a ratio by dividing both parts by the HCF. A rate compares different units.

$12 : 18 = 2 : 3$

$240 in the ratio 5 : 3: 8 parts, 1 part = $30, so **$150** and **$90**

150 km in 2.5 h = **60 km/h**

Best buy: 500 g for $3.60 is **$0.72** per 100 g; 750 g for $5.10 is **$0.68** per 100 g, so the 750 g pack is better value

6 Percentages and money

Per cent means "out of 100". Divide by 100 to get a decimal.

Fraction to %: $\frac{7}{20} \times 100\% = 35\%$ $12.5\% = 0.125$

Find a percentage: 15% of $80 $= 0.15 \times 80 = \$12$

Find the whole: 30% of an amount is $45, so the amount is $45 \div 0.3 = \$150$

Watch out: check by working backwards: 30% of $150 is $45 ✓

7 Percentage increase and decrease

Use a multiplier: 100% plus the increase, or 100% minus the decrease. Percentage change is always out of the original amount.

Up 12%: $50 \times 1.12 = 56$ Down 15%: $80 \times 0.85 = 68$

% change $= \frac{\text{change}}{\text{original}} \times 100\%$: 40 to 50 is $\frac{10}{40} = 25\%$

8 Profits and discounts

Profit = selling price − cost price. Profit % is out of the cost price.

Cost $40, sold for $52: profit $12, $\frac{12}{40} \times 100\% = 30\%$

Mark up $80 by 25%: $80 \times 1.25 = \$100$

20% off $65: $65 \times 0.8 = \$52$

9 Income

Wage: paid per hour. Salary: a set amount a year. Commission: a % of sales.

$28/h: time and a half **$42/h**, double time **$56/h**

Salary $78 000 a year $\div 52 =$ **$1500** a week

4% commission on $15 000 sales = **$600**

10 The PAYG income tax system

Pay As You Go: your employer takes tax out of each pay and sends it to the ATO.

Taxable income = gross income − deductions

Medicare levy: 2%. On $60 000: $0.02 \times 60\,000 = \$1200$

11 Simple interest

The same interest every year, worked out on the amount you started with.

$I = Prt$: principal, rate as a decimal, time in years

$1500 at 4% p.a. for 3 years: $I = 1500 \times 0.04 \times 3 = \$180$

Total $= \$1500 + \$180 = \$1680$

Watch out: 6 months is $t = 0.5$

12 Compound interest and depreciation

Compound: interest is added each year, then earns interest too. Depreciation: value drops by a % each year.

$A = P(1 + r)^n$ $2000 at 6% p.a. for 2 years: $2000 \times 1.06^2 = \$2247.20$

Depreciation $A = P(1 - r)^n$: a $20 000 car losing 15% a year is worth $20\,000 \times 0.85^2 = \$14\,450$ after 2 years

Interest earned = A − P. Here $247.20, more than simple interest ($240).

YEAR 9 MATHS · CHAPTER 2 · MIND MAP

Expressions and linear equations

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1 Algebraic expressions

A pronumeral stands for a number. $3x$ means $3 \times x$.

"5 more than x": $x + 5$ "twice y, minus 4": $2y - 4$

If $a = 3,\ b = -2$: $2a - b^2 = 6 - 4 = 2$

Watch out: substitute negatives in brackets: $(-2)^2 = 4$

2 Simplifying expressions

Only like terms (same letters, same powers) can be added or subtracted.

$4x + 3y - x + 2y = 3x + 5y$

$3a \times 4ab = 12a^2b$ $12xy \div 4x = 3y$

Watch out: $x + x = 2x$ but $x \times x = x^2$

3 Expanding expressions

Multiply every term inside the brackets by the term outside.

$3(x + 4) = 3x + 12$ $x(x + 7) = x^2 + 7x$

$2(x + 3) - (x - 1) = 2x + 6 - x + 1 = x + 7$

Watch out: a minus sign outside changes every sign inside.

4 Solving equations: one side

Undo the operations in reverse order. Do the same to both sides.

$3x + 5 = 20 \Rightarrow 3x = 15 \Rightarrow x = 5$

$\frac{x}{4} - 2 = 3 \Rightarrow \frac{x}{4} = 5 \Rightarrow x = 20$

Check by substituting: $3(5) + 5 = 20$ ✓

5 Brackets and pronumerals on both sides

Expand brackets first, then collect the x terms on one side.

$2(x - 3) = 10 \Rightarrow 2x - 6 = 10 \Rightarrow x = 8$

$5x + 2 = 2x + 14 \Rightarrow 3x = 12 \Rightarrow x = 4$

$4(x + 1) = 2(x + 7) \Rightarrow 4x + 4 = 2x + 14 \Rightarrow x = 5$

Watch out: subtract the smaller x term, so the x stays positive.

6 Solving word problems

Let x be the unknown, write an equation, solve it, then answer in words.

"Triple a number, add 7, get 31": $3x + 7 = 31$, so the number is 8

Rectangle 3 cm longer than it is wide, perimeter 26 cm: $2(w + w + 3) = 26 \Rightarrow w = 5$, so 5 cm by 8 cm

7 Solving linear inequalities

Solve like an equation, but flip the sign when you multiply or divide by a negative.

$2x + 3 < 11 \Rightarrow 2x < 8 \Rightarrow x < 4$

$4 - 2x > 10 \Rightarrow -2x > 6 \Rightarrow x < -3$

On a number line: $<$ or $>$ is an open circle, $\leq$ or $\geq$ is a closed dot.

8 Using formulas

Substitute the values you know, then solve for the one that's left.

$v = u + at$ with $u = 5,\ a = 2,\ t = 3$: $v = 11$

$A = \frac{1}{2}bh$ with $A = 24,\ b = 8$: $24 = 4h$, so $h = 6$

Rearrange for t: $t = \dfrac{v - u}{a}$

9 Simultaneous equations: substitution

Replace one pronumeral with what it equals from the other equation.

$y = 2x$ and $x + y = 12$

$x + 2x = 12 \Rightarrow x = 4$, then $y = 8$

Check in both equations: 8 = 2(4) ✓ and 4 + 8 = 12 ✓

10 Simultaneous equations: elimination

Add or subtract the equations to remove one pronumeral.

$x + y = 10,\ x - y = 4$: add, $2x = 14$, so $x = 7,\ y = 3$

$2x + 3y = 13,\ 2x + y = 7$: subtract, $2y = 6$, so $y = 3,\ x = 2$

11 Applications of simultaneous equations

Two unknowns need two equations. Use a letter for each unknown.

Adult tickets $15, child tickets $8. 7 tickets cost $77.

$a + c = 7$ and $15a + 8c = 77$

Substitute $a = 7 - c$: $105 - 7c = 77$, so $c = 4$ and $a = 3$: 3 adults and 4 children

YEAR 9 MATHS · CHAPTER 3 · MIND MAP

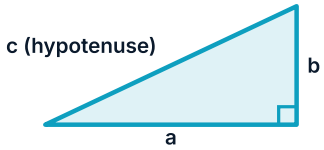
Pythagoras and trigonometry

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1 Pythagoras' theorem

In a right-angled triangle: $c^2 = a^2 + b^2$. c is the hypotenuse, the longest side.

Sides 6 and 8: $c^2 = 36 + 64 = 100$, **so** $c = 10$



2 Finding a shorter side

To find a shorter side, subtract: $a^2 = c^2 - b^2$.

Hypotenuse 13, side 5: $a^2 = 169 - 25 = 144$, **so** $a = 12$

Watch out: a shorter side can never be longer than the hypotenuse.

3 Pythagoras in 2D problems

Draw it, mark the right angle, then find the triangle.

A 5 m ladder reaches 4 m up a wall: the foot is $\sqrt{25 - 16} = 3$ **m out**

Diagonal of a 9 by 12 rectangle: $\sqrt{81 + 144} = 15$

4 Pythagoras in 3D

Find a right triangle on one face first, then use that length in a second triangle.

Box 3 by 4 by 12: base diagonal $\sqrt{9 + 16} = 5$

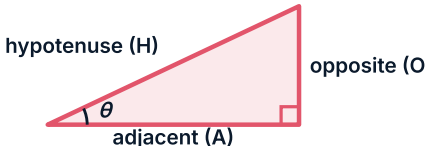
Space diagonal $\sqrt{5^2 + 12^2} = 13$

Shortcut: $d = \sqrt{a^2 + b^2 + c^2}$

5 Trigonometric ratios

SOH CAH TOA. Label the sides from the angle you're using.

$$\sin\theta = \frac{O}{H} \quad \cos\theta = \frac{A}{H} \quad \tan\theta = \frac{O}{A}$$



Watch out: set your calculator to degrees.

6 Finding unknown sides

Pick the ratio that uses the side you know and the side you want.

Opposite 35°, hypotenuse 10: $x = 10\sin 35° \approx 5.74$

Adjacent 8, angle 40°, find opposite: $x = 8\tan 40° \approx 6.71$

7 Solving for the denominator

When x is on the bottom, swap it with the trig value.

$$\cos 30° = \frac{12}{x} \Rightarrow x = \frac{12}{\cos 30°} \approx 13.86$$

Watch out: check the answer makes sense: the hypotenuse must be the longest side.

8 Finding unknown angles

Use the inverse buttons: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.

Opposite 5, hypotenuse 10: $\sin\theta = 0.5$, **so** $\theta = 30°$

Opposite 7, adjacent 4: $\theta = \tan^{-1}(1.75) \approx 60.3°$

9 Applying trigonometry

Angles of elevation (looking up) and depression (looking down) are measured from the horizontal.

20 m from a tree, the angle of elevation to the top is 32°

Height $= 20\tan 32° \approx 12.5$ **m**

Watch out: draw the horizontal line first, then the angle.

10 Bearings

True bearings are measured clockwise from north, with three digits.

North-east is 045°T, south-east 135°T, west 270°T

Walk 10 km on 060°T: east $10\sin 60° \approx 8.66$ **km, north** $10\cos 60° = 5$ **km**

YEAR 9 MATHS · CHAPTER 4 · MIND MAP

Linear relationships

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1 Introduction to linear relationships

A linear rule gives a straight line. y changes by the same amount each step.

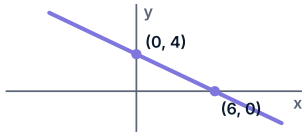
$y = 2x + 1$: $x = 0, 1, 2$ gives $y = 1, 3, 5$

Is $(3, 7)$ on the line? $2(3) + 1 = 7$ ✓ yes

2 Graphing using intercepts

x-intercept: put $y = 0$. y-intercept: put $x = 0$.

$2x + 3y = 12$: x-intercept $(6, 0)$, y-intercept $(0, 4)$



3 Lines with one intercept

Some lines only cross one axis, or pass through the origin.

$y = 3$: horizontal $x = -2$: vertical

$y = 2x$: through the origin $(0, 0)$

4 Gradient

Gradient $m = \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$

$(1, 2)$ to $(3, 8)$: $m = \dfrac{6}{2} = 3$

Watch out: rise over run, not run over rise. Going down to the right is negative.

5 Gradient and direct proportion

$y = kx$: a straight line through the origin. The gradient k is the constant of proportion.

Paid \$25 an hour: $P = 25h$

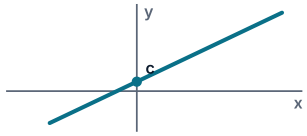
6 hours: $P = 25 \times 6 = \$150$

6 Gradient–intercept form

$y = mx + c$: m is the gradient, c is the y-intercept.

$y = -2x + 5$: $m = -2$, $c = 5$

Rearrange $3x + y = 4$: $y = -3x + 4$



7 Finding the equation of a line

Find m, then substitute a point to find c.

Gradient 2 through $(1, 5)$: $5 = 2(1) + c$, so $y = 2x + 3$

Through $(0, 1)$ and $(2, 7)$: $m = 3$, $c = 1$, so $y = 3x + 1$

8 Midpoint and length

Midpoint: average the coordinates. Length: Pythagoras.

$(2, 3)$ and $(8, 11)$: midpoint $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$

Length $\sqrt{6^2 + 8^2} = 10$

9 Parallel and perpendicular lines

Parallel: same gradient. Perpendicular: gradients multiply to -1.

$y = 2x + 1$ is parallel to $y = 2x - 4$

A line perpendicular to $y = 2x + 1$ has gradient $-\frac{1}{2}$

10 Linear modelling

c is the starting value, m is the rate of change.

Taxi: \$4 flag fall plus \$2.50 per km: $C = 2.5d + 4$

10 km: $C = 2.5 \times 10 + 4 = \$29$

11 Graphical solutions to simultaneous equations

The solution is the point where the two lines cross.

$y = x + 1$ and $y = -x + 5$ cross at $(2, 3)$

Check: $2 + 1 = 3$ ✓ and $-2 + 5 = 3$ ✓

Parallel lines never cross, so there's no solution.

YEAR 9 MATHS · CHAPTER 5 · MIND MAP

Measurement

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1 Errors and accuracy

A measurement is accurate to plus or minus half the smallest unit.

12 cm to the nearest cm: between **11.5 cm** and **12.5 cm**

Greatest possible error: 0.5 cm

2 Length and perimeter

Perimeter is the distance around the outside: add every side.

1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm

Rectangle 8 by 3: $P = 2(8 + 3) = 22$

Watch out: convert to the same unit before you add.

3 Circumference and sectors

$C = \pi d = 2\pi r$. A sector is a fraction $\frac{\theta}{360}$ of a circle.

$r = 5$: $C = 10\pi \approx 31.4$

Sector $r = 6,\ \theta = 90°$: $P = 2(6) + \frac{90}{360} \times 2\pi(6) \approx 21.4$

4 Area

Rectangle lw · triangle $\frac{1}{2}bh$ · parallelogram bh · trapezium $\frac{1}{2}(a+b)h$ · circle πr^2

Trapezium $a = 4,\ b = 10,\ h = 5$: $\frac{1}{2} \times 14 \times 5 = 35$

Circle $r = 3$: $9\pi \approx 28.3$

1 m² = 10 000 cm² (100 × 100).

5 Composite shapes

Split into shapes you know and add, or take a shape away.

10 by 6 rectangle with a semicircle on the 6 side: $60 + \frac{1}{2}\pi(3^2) \approx 74.1$

Watch out: the semicircle's radius is half the side: 3, not 6.

6 Surface area of prisms and pyramids

Add the areas of every face. Drawing the net helps.

Cube with side 4: $6 \times 16 = 96$

Square pyramid, base 6, triangle height 5: $36 + 4 \times \frac{1}{2} \times 6 \times 5 = 96$

7 Surface area of cylinders

$SA = 2\pi r^2 + 2\pi rh$ (two circles plus the curved side).

$r = 3,\ h = 10$: $18\pi + 60\pi = 78\pi \approx 245.0$

An open cylinder (like a cup) has only one circle.

8 Volume of prisms

$V = Ah$: area of the cross-section times the length.

Triangular prism, triangle $\frac{1}{2} \times 4 \times 3 = 6$, length 10: $V = 60$

1 cm³ = 1 mL, 1000 cm³ = 1 L, 1 m³ = 1000 L

9 Volume of cylinders

$V = \pi r^2 h$: the circle's area times the height.

$r = 3$ cm, $h = 10$ cm: $V = 90\pi \approx 282.7$ cm³, about 283 mL

Watch out: use the radius, not the diameter.

YEAR 9 MATHS · CHAPTER 6 · MIND MAP

Indices and surds

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1 Index notation

2^5: 2 is the base, 5 is the index (how many 2s multiply).

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

Prime factors: $72 = 2^3 \times 3^2$

2 Multiplying and dividing

Same base: multiply, add the indices. Divide, subtract them.

$a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$

$x^3 \times x^4 = x^7$ $6a^5 \div 2a^2 = 3a^3$

Watch out: $2^3 \times 2^4 = 2^7$, not 4^7

3 Power of a power, zero index

Power of a power: multiply the indices. Anything to the power 0 is 1.

$(a^m)^n = a^{mn}$: $(x^2)^5 = x^{10}$

$7^0 = 1$ $5x^0 = 5$

4 Brackets and fractions

The power applies to everything inside the brackets.

$(2x^3)^2 = 4x^6$

$\left(\frac{x}{3}\right)^2 = \frac{x^2}{9}$

Watch out: square the 2 too: $(2x)^2 = 4x^2$

5 Negative indices

A negative index means one over: $a^{-n} = \frac{1}{a^n}$.

$2^{-3} = \frac{1}{8}$ $x^{-2} = \frac{1}{x^2}$ $\frac{1}{a^{-2}} = a^2$

6 Scientific notation

Write numbers as $a \times 10^n$ with $1 \leq a < 10$.

$45\,000 = 4.5 \times 10^4$

$0.0062 = 6.2 \times 10^{-3}$

Watch out: big numbers get a positive power, small ones a negative power.

7 Scientific notation and significant figures

Round the first number to the significant figures asked for.

$0.000\,456\,7$ **to 2 s.f.:** 4.6×10^{-4}

$3\,870\,000$ **to 2 s.f.:** 3.9×10^6

$(2.4 \times 10^5) \times (3 \times 10^3) = 7.2 \times 10^8$

8 Fractional indices and surds

$a^{\frac{1}{2}} = \sqrt{a}$, $a^{\frac{1}{3}} = \sqrt[3]{a}$. A surd is a root that isn't a whole number, like $\sqrt{2}$.

$9^{\frac{1}{2}} = 3$ $8^{\frac{1}{3}} = 2$

Simplify: $\sqrt{12} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$

9 Simple operations with surds

Add or subtract like surds only. Multiply: multiply the numbers outside and inside.

$3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$ $\sqrt{3} \times \sqrt{5} = \sqrt{15}$

$2\sqrt{3} \times 4\sqrt{2} = 8\sqrt{6}$

$\sqrt{8} + \sqrt{18} = 2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$

Watch out: $\sqrt{2} + \sqrt{3} \neq \sqrt{5}$

YEAR 9 MATHS · CHAPTER 7 · MIND MAP

Geometry



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1 Angles and triangles

Triangle angles add to 180°. Angles on a straight line add to 180°, around a point 360°.

Two angles are 50° and 60°: the third is **70°**

Exterior angle = sum of the two opposite interior angles

An isosceles triangle has two equal sides and two equal base angles.

2 Parallel lines

Corresponding angles are equal (F), alternate angles are equal (Z), co-interior angles add to 180° (C).

If one angle is 65°, its co-interior angle is **115°**

Watch out: these rules only work when the lines are parallel.

3 Quadrilaterals and polygons

Interior angle sum of an n-sided polygon: $(n-2)\times 180^\circ$.

Quadrilateral: 360°

Hexagon: $4\times 180^\circ = 720^\circ$; each angle of a regular hexagon is 120°

4 Congruent triangles

Congruent means exactly the same shape and size (≅). Tests: SSS, SAS, AAS, RHS.

SSS: all three sides equal

SAS: two sides and the angle between them

RHS: right angle, hypotenuse and one side

Watch out: AAA is not a congruence test: same angles can be different sizes.

5 Using congruence in proof

List the equal parts with a reason for each, name the test, then conclude.

ABCD is a parallelogram. Prove △ABC ≅ △CDA.

AB = CD and BC = DA (opposite sides of a parallelogram)

AC is common, so △ABC ≅ △CDA (SSS)

6 Enlargement and similar figures

Scale factor $= \dfrac{\text{image length}}{\text{original length}}$. Angles stay the same.

A 4 cm side enlarged by a scale factor of 2.5 becomes **10 cm**

Similar figures: same shape, sides in the same ratio.

7 Similar triangles

Tests: AAA, SSS (ratio), SAS (ratio), RHS (ratio). Match the sides in order.

Matching sides 4 and 6: scale factor $\frac{6}{4} = 1.5$

Side 5 matches x: $x = 5\times 1.5 = 7.5$

8 Proving and applying similar triangles

Show the triangles are similar, then use the scale factor.

A 2 m pole casts a 3 m shadow. A tree's shadow is 12 m at the same time.

Height $= 2\times\frac{12}{3} = 8$ m

Watch out: write the vertices in matching order: △ABC ||| △DEF.

YEAR 9 MATHS · CHAPTER 8 · MIND MAP

Algebraic techniques

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1 Expanding binomial products

Multiply every term in the first bracket by every term in the second.

$(x+3)(x+5) = x^2+8x+15$

$(2x-1)(x+4) = 2x^2+7x-4$

2 Perfect squares and DOTS

$(a+b)^2 = a^2+2ab+b^2$ and $(a-b)(a+b) = a^2-b^2$

$(x+4)^2 = x^2+8x+16$

$(x-5)(x+5) = x^2-25$

Watch out: $(x+3)^2 \neq x^2+9$: don't forget the middle term.

3 Factorising expressions

Take out the highest common factor (HCF).

$6x+9 = 3(2x+3)$ $\quad x^2-7x = x(x-7)$

$-4a-8 = -4(a+2)$

Check by expanding: you should get back to the start.

4 Factorising DOTS

Difference of two squares: $a^2-b^2 = (a-b)(a+b)$

$x^2-49 = (x-7)(x+7)$

$4x^2-9 = (2x-3)(2x+3)$

$3x^2-12 = 3(x-2)(x+2)$

5 Factorising by grouping

Group into pairs, take out a factor from each, then the common bracket.

$x^2+3x+2x+6 = x(x+3)+2(x+3)$

$= (x+3)(x+2)$

6 Monic quadratic trinomials

x^2+bx+c: find two numbers that multiply to c and add to b.

$x^2+7x+12 = (x+3)(x+4)$

$x^2-x-6 = (x-3)(x+2)$

7 Trinomials ax^2+bx+c

Find two numbers that multiply to $a \times c$ and add to b. Split the middle term, then group.

$2x^2+7x+3$: $2 \times 3 = 6$, use 6 and 1

$2x^2+6x+x+3 = 2x(x+3)+1(x+3)$

$= (2x+1)(x+3)$

8 Algebraic fractions: × and ÷

Factorise, cancel common factors, then multiply. To divide, flip the second fraction.

$\frac{x+2}{6} \times \frac{9}{x+2} = \frac{9}{6} = \frac{3}{2}$

$\frac{4x}{5} \div \frac{2x}{15} = \frac{4x}{5} \times \frac{15}{2x} = 6$

9 Algebraic fractions: + and −

Use a common denominator, then add or subtract the numerators.

$\frac{x}{3}+\frac{x}{4} = \frac{4x}{12}+\frac{3x}{12} = \frac{7x}{12}$

$\frac{2}{x}-\frac{1}{2x} = \frac{4}{2x}-\frac{1}{2x} = \frac{3}{2x}$

10 Further simplification

Factorise first, then cancel whole factors.

$\frac{x^2-9}{x+3} = x-3$

$\frac{x^2+5x+6}{x+2} = x+3$

Watch out: cancel factors, never single terms: $\frac{x+3}{3} \neq x$

11 Equations with algebraic fractions

Multiply both sides by the lowest common denominator to clear the fractions.

$\frac{x+1}{3} = \frac{x-2}{2} \Rightarrow 2(x+1) = 3(x-2) \Rightarrow x = 8$

$\frac{x}{2}+\frac{x}{3} = 10$: multiply by 6, $3x+2x = 60$, so $x = 12$

YEAR 9 MATHS · CHAPTER 9 · MIND MAP

Probability and statistics

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1 Review of probability

$P(\text{event}) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$, from 0 to 1.

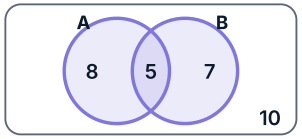
One die: $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$

$P(\text{not } A) = 1 - P(A)$: $P(\text{not } 6) = \frac{5}{6}$

2 Venn diagrams and two-way tables

Fill in the overlap first.

30 students: $P(A) = \frac{13}{30}$, $P(A \cap B) = \frac{1}{6}$



3 Set notation

$A \cap B$: A and B. $A \cup B$: A or B (or both). A': not A.

From the Venn: $n(A \cup B) = 8 + 5 + 7 = 20$

$n(A') = 7 + 10 = 17$

4 Arrays for two-step experiments

List every pair of outcomes in a table.

Two dice: $6 \times 6 = 36$ outcomes

Sum of 7: 6 ways, so $P = \frac{6}{36} = \frac{1}{6}$

5 Tree diagrams

Multiply along the branches. Add the results you want.

Two coins: $P(\text{HH}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$P(\text{at least one head}) = 1 - \frac{1}{4} = \frac{3}{4}$

3 red and 2 blue, take 2 without replacement: $P(\text{both red}) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$

Watch out: without replacement, the second branch has one fewer.

6 Relative frequency

Relative frequency $= \dfrac{\text{times it happened}}{\text{number of trials}}$

A pin lands point-up 36 times in 120 throws: $\frac{36}{120} = 0.3$

Expected in 500 throws: $0.3 \times 500 = 150$

More trials give a better estimate.

7 Data and sampling

A sample should represent the whole population. Random samples avoid bias.

Categorical: colours, types. Numerical: discrete (counts) or continuous (measurements)

Asking only gym members how often people exercise is biased

8 Mean, median and mode

Mean: add and divide. Median: middle value in order. Mode: most common.

3, 5, 5, 8, 9: mean $\frac{30}{5} = 6$, median 5, mode 5

2, 4, 7, 9: median $\frac{4+7}{2} = 5.5$

9 Stem-and-leaf plots

The stem is the tens, the leaf is the units. Leaves go in order.

2 | 1 4 7 and 3 | 0 2 (key 2 | 4 = 24)

Data 21, 24, 27, 30, 32: median **27**

10 Grouping data into classes

Group numerical data into equal class intervals, then draw a histogram.

Classes 0–9, 10–19, 20–29 with a frequency for each

Class centre of 10–19: **14.5**

With grouped data, use the class centres to estimate the mean.

11 Range and interquartile range

Range = max − min. IQR $= Q_3 - Q_1$, the spread of the middle half.

2, 4, 5, 7, 8, 10, 12, 15: range $15 - 2 = 13$

$Q_1 = 4.5$, median 7.5, $Q_3 = 11$, so IQR $= 6.5$

12 Box plots

Draw the five-number summary: min, Q_1, median, Q_3, max. The box holds the middle 50%.

The data from card 11:



YEAR 9 MATHS · CHAPTER 10 · MIND MAP

Quadratic equations and graphs

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1 Quadratic equations

Null Factor Law: if $ab = 0$, then $a = 0$ or $b = 0$.

$(x - 3)(x + 5) = 0$

$x = 3$ or $x = -5$

A quadratic can have two, one or no solutions.

2 Solving $ax^2 + bx = 0$ and $x^2 = d$

Take out the common factor. For $x^2 = d$, remember $\pm$.

$2x^2 - 6x = 0 \Rightarrow 2x(x - 3) = 0 \Rightarrow x = 0$ or 3

$x^2 = 16 \Rightarrow x = \pm 4$

Watch out: don't divide by x: you'd lose $x = 0$.

3 Solving $x^2 + bx + c = 0$

Make one side 0, factorise, then use the Null Factor Law.

$x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0$, so $x = -4$ or 3

$x^2 = 5x - 6 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 2$ or 3

4 Using quadratics to solve problems

Write an equation, solve it, then reject answers that don't make sense.

A rectangle is 3 m longer than it is wide, area 40 m²: $x(x + 3) = 40$

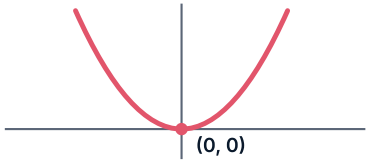
$x^2 + 3x - 40 = 0 \Rightarrow (x + 8)(x - 5) = 0$

Width can't be -8, so it's 5 m by 8 m

5 The parabola

$y = x^2$ is a parabola. It has a turning point and an axis of symmetry.

$a > 0$: opens up (minimum). $a < 0$: opens down (maximum).

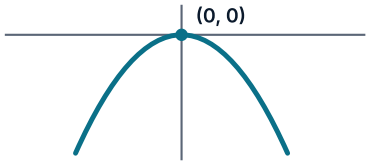


6 Dilations and reflections

$y = ax^2$: a bigger $|a|$ makes it narrower; a negative a flips it upside down.

$y = 3x^2$: narrower $y = \frac{1}{2}x^2$: wider

$y = -x^2$: reflected in the x-axis



7 Translations

$y = (x - h)^2 + k$ has turning point (h, k).

$y = (x - 2)^2 + 3$: turning point $(2, 3)$

$y = (x + 1)^2 - 4$: turning point $(-1, -4)$

Watch out: the sign of h flips: $(x + 1)$ moves left 1.

8 Intercept form

$y = (x - a)(x - b)$ crosses the x-axis at a and b. The turning point is halfway between.

$y = (x - 1)(x - 5)$: x-intercepts 1 and 5, axis $x = 3$

Turning point $y = (3 - 1)(3 - 5) = -4$, so $(3, -4)$; y-intercept $(0, 5)$

